

# Introduction to Road Network Planning and Navigation

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2. Introduction to Road Network Planning

## 1. Course Content

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Understand the road network planning and navigation function. This section provides a brief overview of the basic concepts, functions, and application scenarios of "road network planning and navigation."

## 2. Introduction to Road Network Planning

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- The core of the road network planning and navigation function is the use of Nav2's route server to help users create, edit, and manage route maps for robot navigation. A route map represents the effective path a robot can follow in its environment, consisting of nodes (path points) and edges (connections between path points). Unlike ordinary free-space planning, path-based navigation ensures that the robot follows a specific, predefined path.

Road network planning and navigation is commonly used in the following scenarios:

- Industrial environments where specific routes must be followed
  - Application example: Factory AGV robots transport parts along a pre-set road network, avoiding production lines, personnel passages, and heavy equipment areas.
- Warehouse operations requiring structured movement patterns
  - Application example: Parts warehouses, where the road network is divided into zones according to material type. Robots access parts along fixed routes, ensuring first-in, first-out (FIFO) while avoiding shelf collisions.
- Facilities with restricted areas or priority routes
- Large outdoor urban or natural environments
  - Application case: Route planning in navigation software is based on urban lane networks; autonomous sightseeing vehicles in scenic areas only travel along the park's fixed road network, avoiding ecological protection zones and dangerous sections.